Toyin Onwunaru

Clinical AI Evaluation & Applied ML

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SUMMARY

Penultimate-year medical student at the University of Port Harcourt who evaluates clinical AI the way a clinician works up a patient: look for what does not fit, and say what is still uncertain. Ran the literature review for the EU MDR update of the Thermalytix clinical evaluation at Niramai Health Analytix (AI breast cancer screening), taking 513 database records down to 5 eligible studies. Builds Python tools that test clinical AI for the errors that matter and flag uncertainty instead of guessing.

EXPERIENCE

Clinical Evaluation Intern, Literature Review 2026

Niramai Health Analytix, Bangalore, India (remote). AI breast cancer screening company; device under evaluation: Thermalytix.

- Searched PubMed (25 strings), the Cochrane Library, ClinicalTrials.gov and Europe PMC for 2020 to 2026: 513 records identified, 260 duplicates removed, and a repeatable search log written for a reviewer to rerun.
- Screened 248 records on title and abstract and assessed 23 reports at full text against the existing exclusion criteria. Added three codes where the old ones fell short (no objective endpoint, not peer reviewed or no full text, duplicate cohort). Result: 5 studies included, with every exclusion carrying a code and a one sentence reason.
- Extracted sensitivity, specificity, PPV, NPV, cancer detection rate and recall rate for each included study and scored them on the team's 100 point appraisal grid, writing out the reason for every grade.
- Checked my own screening: re-read a random sample of 10 exclusions, read 10 borderline excluded papers in full and recoded 3. Found that most thermography AI papers I read in full were tested on one public image database rather than on prospectively recruited patients.
- Built the PRISMA flow, the methods section and a limitations list in which every count reconciles to the workbook.

SEO Copywriter & Long-Form Content Strategist, Freelance (Upwork) Alongside medical training

100% Job Success, 21 jobs completed for clients on several continents. The same habit of testing a claim against evidence and writing it up clearly, now pointed at clinical AI.

PROJECTS

Clinical Note Eval Generator Python, rule-based error injection, eval dataset design

Generates clinical notes with labeled, injected errors in six categories (dosage, allergy conflict, contradictory vitals, drug interaction, unit error, dropped follow-up), mixed with clean notes, to score whether a clinical AI catches what a clinician would and stays quiet when nothing is wrong. Live demo

Lab-Value-to-Plain-English Explainer Python, rule-based clinical reasoning, confidence scoring

Interprets five common lab tests and separates its confidence from its verdict, naming the missing context (sex, build, comorbidities) whenever a result is ambiguous. Live demo

NaijaTriageAI Flask, Supabase, offline-first PWA

Installable triage tool for chest pain and maternal and neonatal emergencies (PPH, neonatal asphyxia) using Bayesian models calibrated to Nigerian disease prevalence. GitHub

Sickle Cell Disease NLP Phenotyping Pipeline Bio+ClinicalBERT, UMAP, HDBSCAN

Extracts structured, clustered SCD phenotypes from free-text clinical narrative in the PMC-Patients corpus, with GradientBoostingClassifier for downstream classification. GitHub

EDUCATION

MBBS, University of Port Harcourt, Nigeria Penultimate year of a 6-year programme

Clinical rotations in Obstetrics and Gynaecology and Paediatrics.

SKILLS

Clinical evidence: systematic literature review (PRISMA), screening and exclusion criteria design, diagnostic accuracy appraisal, data extraction, evidence writing for EU MDR clinical evaluation

AI evaluation: prompt engineering and LLM evaluation design, eval dataset design, error injection, confidence scoring

Technical: Python, SQL, Django, Flask, Supabase, FHIR and HL7, EHR data literacy, Bio+ClinicalBERT, UMAP, HDBSCAN, GradientBoostingClassifier